1. RtHDDump Token Inference Report

1.1. Abstract

This report summarizes 27 RtHDDump captures in this repository by interpreting filename tokens that encode device state. The inferred variables include preferred device, headset plug state, Dolby mode, and system audio enhancement flags. Counts and proportions are reported with charts and tables to make the filename changes easy to interpret at a glance.

1.2. Data and methods

- Source data: `rthddump_summary.csv` generated by `rthddump_report.py`.
- Tokens and interpretations:
  - `spk` = speaker, `h` = headset.
  - First device token = preferred device.
  - `dual` = headset plugged (absence = unplugged).
  - `dolby`/`dolbys` = Dolby for speaker, `dolbyh` = Dolby for headset.
  - `enhance` after `spk` = speaker system audio enhancement, `enhance` after `h` = headset system audio enhancement.
- Statistics: simple counts and proportions across all dumps.

1.2.1. Inferred token dictionary (final interpretation)

Token	Meaning
`spk`	Speaker device path
`h`	Headset device path
`dual`	Headset plugged in (absence means unplugged)
`dolby`, `dolbys`	Dolby enabled for speaker
`dolbyh`	Dolby enabled for headset
`enhance` after `spk`	Speaker system audio enhancement enabled
`enhance` after `h`	Headset system audio enhancement enabled

1.3. Results

1.3.1. Dataset overview

Metric	Count	Share
Total RtHDDump files	27	100%
Preferred device = speaker	17	63.0%
Preferred device = headset	10	37.0%
Headset plugged	23	85.2%
Headset unplugged	4	14.8%

Speaker	17
Headset	10

Table 1: Preferred device distribution

Plugged (dual)		23
Unplugged		4

Table 2: Headset plug state

1.3.2. Feature enablement

Feature	Enabled	Disabled	Enable rate
Speaker Dolby	18	9	66.7%
Speaker system audio enhancement	19	8	70.4%
Headset Dolby	8	19	29.6%
Headset system audio enhancement	17	10	63.0%

Speaker Dolby		18
Speaker enhancement		19
Headset Dolby		8
Headset enhancement		17

Table 3: Feature enablement counts

1.3.3. Data quality checks

Check	Result
Files with unknown filename tokens	0
Files with recorded SHA-256 in CSV	27

1.4. Discussion

Most dumps favor the speaker as the preferred device, while the headset is plugged in for the majority of captures. Speaker Dolby and speaker enhancement appear more frequently than the corresponding headset features, suggesting the speaker path is typically configured with additional processing in this dataset. Headset Dolby is present in fewer than one-third of dumps, while headset enhancement appears in roughly two-thirds, indicating distinct tuning profiles depending on the device path.

1.5. Conclusion

The filename tokens consistently encode device selection and processing features. The dataset is dominated by speaker-preferred configurations with headset plugged in, and speaker processing features appear more frequently than headset Dolby. The full per-file inference table below provides the definitive mapping from each dump to its inferred device state.

1.6. Appendix A: Per-file inference results

File	Preferred	Headset plugged	Speaker Dolby	Speaker Enhance	Headset Dolby	Headset Enhance
RtHDDump_dual_hs_dolby_enhance_spk_dolby.txt	headset	True	True	False	False	True
RtHDDump_dual_hs_dolby_enhance_spk_dolby_enhance.txt	headset	True	True	True	False	True
RtHDDump_dual_hs_dolby_enhance_spk_enhance.txt	headset	True	True	True	False	True
RtHDDump_dual_hs_dolby_spk_dolby.txt	headset	True	True	False	False	False
RtHDDump_dual_hs_dolby_spk_dolby_enhance.txt	headset	True	True	True	False	False

RtHDDump_dual_headset_dolby_spk_enhance.txt	headset	True	True	True	False	False
RtHDDump_dual_headset_enhance_spk_dolby.txt	headset	True	True	False	False	True
RtHDDump_dual_headset_enhance_spk_dolby_enhance.txt	headset	True	True	True	False	True
RtHDDump_dual_headset_enhance_spk_enhance.txt	headset	True	False	True	False	True
RtHDDump_dual_headset_spk_dolby_enhance.txt	headset	True	True	True	False	False
RtHDDump_dual_spk_dolbyh_enhance_h_dolbyh_enhance.txt	speaker	True	False	True	True	True
RtHDDump_dual_spk_dolbyh_enhance_h_enhance.txt	speaker	True	False	True	True	True
RtHDDump_dual_spk_dolbyh_h_dolbyh_enhance.txt	speaker	True	False	False	True	True
RtHDDump_dual_spk_dolbys_enhance_h_dolbyh.txt	speaker	True	True	True	True	False
RtHDDump_dual_spk_dolbys_enhance_h_dolbyh_enhance.txt	speaker	True	True	True	True	True
RtHDDump_dual_spk_dolbys_enhance_h_dolbys.txt	speaker	True	True	True	False	False
RtHDDump_dual_spk_dolbys_enhance_h_dolbys_enhance.txt	speaker	True	True	True	False	True
RtHDDump_dual_spk_dolbys_enhance_h_enhance.txt	speaker	True	True	True	False	True
RtHDDump_dual_spk_dolbys_h_dolbyh_enhance.txt	speaker	True	True	False	True	True
RtHDDump_dual_spk_enhance_h_dolbyh_enhance.txt	speaker	True	False	True	True	True
RtHDDump_dual_spk_enhance_h_dolbys_enhance.txt	speaker	True	True	True	False	True
RtHDDump_dual_spk_enhance_h_enhance.txt	speaker	True	False	True	False	True
RtHDDump_dual_spk_h_dolbyh_enhance.txt	speaker	True	False	False	True	True
RtHDDump_spk.txt	speaker	False	False	False	False	False
RtHDDump_spk_dolby.txt	speaker	False	True	False	False	False
RtHDDump_spk_dolby_enhance.txt	speaker	False	True	True	False	False
RtHDDump_spk_enhance.txt	speaker	False	False	True	False	False

1.7. Appendix B: Regeneration

Generate the CSV summary and the PDF report from the current dumps:

```
python rthddump_report.py --output rthddump_summary.csv
typst compile RtHDDump_Report.typ RtHDDump_Report.pdf
```